

Elise Boisson

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Aug. 2024 – May 2027

Bachelor of Science, Mechanical Engineering | GPA: 3.84

Denmark Technical University, Kongens Lyngby, Denmark (Study Abroad)

Jan. 2026 – June 2026

Elective Courses: Marine and Ocean Engineering, Fundamentals of Aerodynamics, Green Fuels and Power to X

Current Coursework: Finite Element Analysis, Intro to Spaceflight Mechanics, System Dynamics, Heat & Fluid Transfer Lab

PROJECTS

Mars Rotorcraft Blade Optimization | Fund. of Aerodynamics - Group Project

Spring 2026

- Developed a Python-based aerodynamic algorithm using a brute-force parameter sweep to optimize blade twist, chord length, and airfoil profile, minimizing hover thrust based on NASA's Ingenuity helicopter
- Generated 3D coordinate data from optimized airfoil geometry and led CAD modeling in SOLIDWORKS

Animatronic Axolotl Tail Design | Cornell's Entertainment, Eng. and Design Club - Individual Contribution

Spring 2024

- Designed a two-axis (side-to-side and vertical) animatronic tail mechanism in Fusion 360, using octopus tentacle biomechanics as inspiration for a linked-member structure
- Engineered string-and-gear-motor actuation system with integrated routing holes; validated design through 3D-printed prototype and tested the degree of motion of the design with string motion

Torque-Measuring Wrench Design | Mechanics of Materials (MAE 3270) - Final Project

Fall 2025

- Designed a torque-measuring wrench meeting strict safety factors ($FS \geq 4$ yield, ≥ 2 crack growth, ≥ 1.5 fatigue) and strain gauge output ≥ 1 mV/V, validated via hand calculations and MATLAB automation
- Performed ANSYS FEA across multiple mesh densities and boundary conditions, selecting Aluminum-2024-T3 and a clamped-fillet geometry for optimal structural performance

WORK EXPERIENCE

Academic Excellence Workshop for General Chemistry, Facilitator

Aug. 2025 – Present

- Leading a two-hour review session on general chemistry for a class of 20 students
- Collaborating with a co-facilitator to foster a supportive and engaging learning environment
- Creating instructional materials, including slides and worksheets, to enhance student understanding

Engineering Thermodynamics Grader

Aug. 2026 – Present

- Graded homework and exams for an introductory thermodynamics course, assessing problem-solving methodology
- Created and applied consistent rubrics to ensure fair, accurate evaluation across a cohort of 230+ students

RESEARCH EXPERIENCE

The Rockefeller University, PI Prof. Jean-Laurent Casanova, HHMI Research Intern

June 2025 – Aug. 2025

- Participated in HHMI's Summer Undergraduate Research Program
- Mapped the functional elements of 3'-untranslated regions in just under 20k MANE transcripts
- Filtered and organized over 100k entries in databases on genetic features using Python via Jupyter Notebooks
- Presented weekly slide updates to mentor and attended lab meetings to stay informed on ongoing projects
- Synthesized 9 weeks of research into a 5-minute presentation to students and mentors in related fields

Weill Cornell Medicine, PI Prof. Carl Nathan, Research Intern

July – Sep. 2022, 2023

- Conducted protein purifications and enzyme kinetic experiments on proteins from drug-resistant bacteria
- Tested buffer and reaction conditions to maximize substrate production in enzyme kinetic experiments
- Wrote & submitted a research paper to Terra NYC STEM Fair 2024; won 1st place in Microbiology cohort

TECHNICAL SKILLS

CAD & Design: Fusion 360, SolidWorks

Programming & Analysis: Python, MATLAB, ANSYS, Java, Excel, Prism

Fabrication: Manual mill operation, lathe operation, 3D printing

Languages: French (Fluent)

CLUBS

Cornell's Entertainment, Engineering and Design Club (CEED-C) | Knit with Care | Cornell's Gymnastics Club